$M = 0.2(2\times 16.0\ g/mol) + 0.8(2\times 14.0\ g/mol) = 28.8\ g/mol$
$= 2.88\times10^{-2}\ kg/mol$

Both gases are diatomic, so $\gamma = 7/5$

$T = 273 + 20 = 293\ K$

$$\Rightarrow v = \sqrt{\frac{(7/5)(8.314\ J/mol\cdot K)(293\ K)}{(2.88\times10^{-2}\ kg/mol)}} = \boxed{344\ m/s}$$

Let's now look at particle displacement and pressure as sinusoids in longitudinal waves. Start with displacement:

$$s(x,t) = A\cos(kx - \omega t + \phi_0) \qquad A = \text{displacement amplitude}$$

From the definition of the bulk modulus:

$$\Delta P = -B\frac{\Delta V}{V}$$



small volume element

$\Delta V = A\,\Delta s$

$V = A\,dx$

$$\Rightarrow \Delta P = -B\frac{A\,\Delta s}{A\,dx}$$ take infinitesimal limit $\frac{\Delta s}{dx} \to \frac{\partial s}{\partial x}$

$$\Delta p(x,t) = -B\frac{\partial s}{\partial x} = -B\left(-kA\sin(kx-\omega t+\phi_0)\right)$$

The pressure wave is thus given by

$$\boxed{\Delta p(x,t) = BkA\sin(kx - \omega t + \phi_0)}$$

max pressure change $\Delta P_{max} = BkA$

and the displacement wave is given by

$$\boxed{s(x,t) = A\cos(kx - \omega t + \phi_0)}$$

These are 90° out of phase

$\Rightarrow$ simulation on slide shows why.

* Important distinction

Δp is the difference from ambient pressure, just like Δs is the displacement from equilibrium.

The pressure (absolute) as a function of x,t is

$$P(x,t) = P_0 + \Delta p(x,t)$$

P_0 = equilibrium pressure.

As a wave (longitudinal or transverse) passes through a medium, energy is transported: the elements of the medium carry both kinetic and potential energy as they undergo simple harmonic motion.

Start with

$$y(x,t) = A\cos(kx - \omega t + \phi_0)$$
$$v_y(x,t) = A\omega\sin(kx - \omega t + \phi_0)$$

Consider an element of the medium with mass dm and length dx. As the wave passes through, the element has kinetic energy

$$dK = \frac{1}{2}\,dm\,(v_y)^2 = \begin{cases} \frac{1}{2}\mu\,dx\,(A\omega)^2\sin^2(kx-\omega t+\phi_0) \\ \text{for transverse on a string} \\ \underline{\text{or}} \\ \frac{1}{2}\rho S\,dx\,(A\omega)^2\sin^2(kx-\omega t+\phi_0) \\ \text{for longitudinal waves} \end{cases}$$

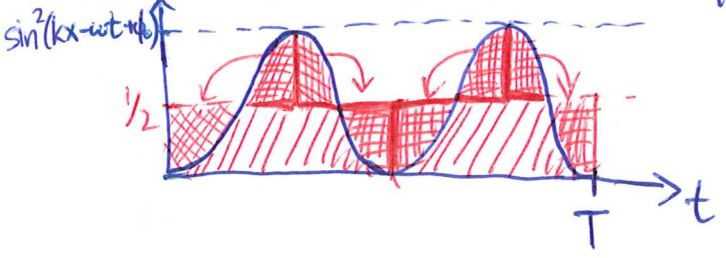
Here S is the cross-sectional area of the medium the wave is travelling through. This will be important when we look at <u>intensity</u> for longitudinal waves. For now we will focus on transverse waves.

The <u>rate</u> at which kinetic energy flows into and out of the element is

$$\frac{dK}{dt} = \frac{1}{2}\mu\frac{dx}{dt}(A\omega)^2\sin^2(kx-\omega t+\phi_0) \qquad \text{but } \frac{dx}{dt} \equiv v$$

$$\Rightarrow \frac{dK}{dt} = \frac{1}{2}\mu v\,(A\omega)^2\sin^2(kx-\omega t+\phi_0)$$

Note this is time dependent because it depends on what moment of the periodic cycle we are talking about. A more useful quantity is the time-average $\left\langle \frac{dK}{dt} \right\rangle$ over a complete cycle. The only time dependence is from the $\sin^2(kx-\omega t+\phi)$ piece:



the average of the square of a sinusoid over one cycle is $\frac{1}{2}$

$$\Rightarrow \left\langle \frac{dK}{dt} \right\rangle = \frac{1}{2}\mu v \omega^2 A^2 \left(\frac{1}{2}\right)$$

Potential energy is also transmitted and in SHM $\langle K \rangle = \langle U \rangle$ so (see Extra Notes 4 for proof)

$$\left\langle \frac{dU}{dt} \right\rangle = \frac{1}{2}\mu v \omega^2 A^2 \left(\frac{1}{2}\right) \quad \text{as well}$$

The <u>total energy</u> is $E = K + U$ and the power is

$$\langle P \rangle = \left\langle \frac{dK}{dt} \right\rangle + \left\langle \frac{dU}{dt} \right\rangle = \boxed{\frac{1}{2}\mu v \omega^2 A^2} \quad \text{using } v = \sqrt{\frac{T}{\mu}}$$

<u>or</u> $$\boxed{\langle P \rangle = \frac{1}{2}\frac{T}{v}\omega^2 A^2}$$

<u>or</u> $$\boxed{\langle P \rangle = \frac{1}{2}\sqrt{T\mu}\,\omega^2 A^2}$$

5 minute break

This last form is interesting because of the quantity $\sqrt{T\mu} = \frac{T}{v}$, which we call the mechanical impedance of the string. It is a measure of how large the perturbation (amplitude of the wave) will be, given the forces responsible. A high mechanical impedance means low amplitude waves, and vice versa.

$\boxed{Z = \sqrt{T\mu}}$ for transverse waves on a string.

* See Extra Notes 4 for a more complete derivation. I am skipping it to save time.

$\boxed{\langle P \rangle = \frac{1}{2} Z \omega^2 A^2}$

What about longitudinal waves in a medium? Acoustic impedance

Solid metal: $\boxed{Z = \sqrt{Y\rho}}$ and $\boxed{\langle P \rangle = \frac{1}{2} Z S \omega^2 A^2}$

S: cross-sectional area of the rod

Fluid: $\boxed{Z = \sqrt{B\rho}}$ and $\boxed{\langle I \rangle = \frac{\langle P \rangle}{S} = \frac{1}{2} Z \omega^2 A^2}$

$\langle I \rangle$: intensity

"plane waves"

In a fluid, it is more useful to define the intensity of the longitudinal wave passing through, which is the power per cross-sectional area.

For "plane waves", the energy is not spreading out, so the intensity remains constant. This is not true for spherical waves produced by a point source. We will see how to account for this on Thursday.